1. Considere el modelo jerarquico de la forma:

$y_i|\lambda \overset{iid}{\sim} Exp(\lambda)$, i.e $p(y_i|\lambda) = \frac{1}{\lambda}\exp\left(\frac{-y_i}{\lambda}\right)$ para $y_i \geq 0$, $\lambda > 0$, $i = 1, \cdots, n$

$\lambda \sim Gam\text{-}inv(\alpha, \beta)$, i.e $p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{-(\alpha+1)}\exp\left(\frac{-\beta}{\lambda}\right)$ $\alpha > 0$ y $\beta > 0$

a. Hallar la distribución predictiva posterior de una observación futura y^*

$$p(y^*|y) = \int_\Theta p(y^*|\lambda)\, p(\lambda|y)\, d\lambda$$

log-verc.

$$p(\lambda|y) \propto \prod_{i=1}^n \left(\frac{1}{\lambda}\exp\left(\frac{-y_i}{\lambda}\right)\right) \cdot p(\lambda)$$

$$= \lambda^{-n}\exp\left(\frac{\sum -y_i}{\lambda}\right) \cdot \frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{-(\alpha+1)}\exp\left(\frac{-\beta}{\lambda}\right)$$

$$= \lambda^{-n-\alpha-1}\exp\left(\frac{-\sum y_i - \beta}{\lambda}\right) \quad \lambda|y \sim \text{Gamma inv}\left(\alpha+n, \sum_{i=1}^n y_i + \beta\right)$$

$$p(y^*|y) = \int_\Theta \frac{1}{\lambda}\exp\left(\frac{-y^*}{\lambda}\right) \cdot \frac{\left(\sum_{i=1}^n y_i + \beta\right)^{\alpha+n}}{\Gamma(\alpha+n)} \lambda^{-(n+\alpha+1)}\exp\left(\frac{-\left(\sum y_i + \beta\right)}{\lambda}\right) d\lambda$$

$$\frac{\left(\sum_{i=1}^n y_i + \beta\right)^{\alpha+n}}{\Gamma(\alpha+n)} \int_\Theta \lambda^{-(n+\alpha+1+1)}\exp\left(\frac{-\left(\sum_{i=1}^n y_i + \beta + y^*\right)}{\lambda}\right) d\lambda$$

Normalización constante de inv-gamma.

$$= \frac{\left(\sum y_i + \beta\right)^{\alpha+n}}{\Gamma(\alpha+n)} \frac{\Gamma(\alpha+n+1)}{\left(\sum_{i=1}^n y_i + \beta + y^*\right)^{\alpha+n+1}}$$

Depende de datos e hiperparametros.

b. Previa de Jeffreys de λ $\quad I(\theta)$ = Información de Fisher. $\quad nI = I_n$

$$p_J(\lambda) \propto \sqrt{I(\lambda)} \qquad I(\lambda) = -E_{y|\lambda}\left[\frac{d^2}{d\lambda}\log p(y|\lambda)\right]$$

$$p(y|\lambda) = \lambda^{-n}\exp\left(\frac{\sum -y_i}{\lambda}\right) \qquad \log p(y|\lambda) = -n\log(\lambda) - \frac{\sum y_i}{\lambda} \qquad \frac{d}{d\lambda}\log p(y|\lambda) = \frac{-n}{\lambda} + \frac{\sum y_i}{\lambda^2}$$

$$\frac{d^2}{d\lambda}\log p(y|\lambda) = \frac{n}{\lambda^2} - \frac{2\sum y_i}{\lambda^3} \qquad -E_{y|\lambda}\left[\frac{d^2}{d\lambda}\log p(y|\lambda)\right] = \frac{-n}{\lambda^2} + \frac{2n\lambda}{\lambda^3}$$

$$n\left(\frac{-1}{\lambda^2} + \frac{2}{\lambda^2}\right) \qquad p_J(\lambda) \propto \sqrt{\frac{1}{\lambda^2}} = \frac{1}{\lambda} \qquad E[y_i] = \lambda$$

Cambios de *

$$p(y^*|y) = \int_\theta p(y^*, \theta|y)\, d\theta = \int_\theta p(y^*|\theta, y)\, p(\theta|y)\, d\theta$$
$$= \int_\theta p(y^*|\theta)\, p(\theta|y)\, d\theta.$$

2. Sea X una variable aleatoria con distribución Gamma con parametro de forma α y parametro de razón β. Hallar la función de densidad de $Y = 1/X$

$X \sim \text{Gamma}(\alpha, \beta)$ Teorema de transformación, la distribución de y esta dada por:

$$f_Y(y) = f_X(g^{-1}(y)) \left| J \right| \quad \text{Si } Y = 1/X \quad X = \frac{1}{y} = g^{-1}(y)$$

$$= \frac{\beta^\alpha}{\Gamma(\alpha)} \left(\frac{1}{y}\right)^{\alpha-1} e^{-\beta/y} \cdot \frac{1}{y^2} = \frac{\beta^\alpha}{\Gamma(\alpha)} \cdot \frac{1}{y^{\alpha-1+2}} e^{-\beta/y}$$

$$= \frac{\beta^\alpha}{\Gamma(\alpha)} y^{-(\alpha+1)} e^{-\beta/y}, \quad y > 0$$

$y \sim \text{Gamma-Inversa}(\alpha, \beta)$.

3. El estimador optimo del parámetro $\theta \in \Theta \subseteq R$ de acuerdo con la regla de bayes se define como el estimador $\tilde{\theta} = \tilde{\theta}(y)$ que minimiza la perdida esperada posterior dada por:

$$E_{\theta|y}(L(\theta, \tilde{\theta})) = \int_\theta L(\theta, \tilde{\theta}) P(\theta|y) d\theta$$

donde $L(\theta, \tilde{\theta})$ es una función de perdida (Estimar θ por medio $\tilde{\theta}$) y $y = (y_1, \cdots, y_n)$ es un conjunto de datos observado. Muestre que si $L(\theta, \tilde{\theta}) = (\theta - \tilde{\theta})^2$, entonces el estimador óptimo de acuerdo con la regla de bayes es la media posterior $\tilde{\theta} = E(\theta|y)$

4. Considere el modelo $X|\theta \sim \text{Binomial}(m,\theta)$ con $\theta \sim \text{Beta}(a,B)$, donde a y B son los hiperparametros del modelo. Se $y|\theta \sim \text{Binomial}(n,\theta)$. Hallar la distribución predictiva posterior $P(y|x)$, asumiendo que y y X son condicionalmente independientes dado θ.

$$P(y|x) = \int_\theta P(y|\theta)\, P(\theta|x)\, d\theta.$$

$$P(\theta|x) \propto P(x|\theta)\, P(\theta) = \binom{m}{x} \theta^x (1-\theta)^{m-x} \frac{\Gamma(a+B)}{\Gamma(a)\Gamma(B)} \cdot \theta^{a-1} (1-\theta)^{B-1}$$

$$\propto \theta^{a+x-1} (1-\theta)^{(m-x+B)-1}$$

Nucleo de una beta.

$$\theta|x \sim \text{Beta}(a+x,\ m-x+B)$$

$$P(y|x) = \int_\theta \binom{n}{y} \theta^y (1-\theta)^{n-y} \frac{\Gamma(a+m+B)}{\Gamma(a+x)\Gamma(m-x+B)} \theta^{a+x-1} (1-\theta)^{m-x+B-1}\, d\theta.$$

$$= \frac{\Gamma(a+m+B)}{\Gamma(a+x)\Gamma(m-x+B)} \binom{n}{y} \int_\theta \theta^{(a+x+y)-1} (1-\theta)^{(m-x+B+n-y)-1}\, d\theta$$

Constante de normalización

$$= \frac{\Gamma(n+1)}{\Gamma(y+1)\Gamma(n-y+1)} \frac{\Gamma(a+m+B)}{\Gamma(a+x)\Gamma(m-x+B)} \cdot \frac{\Gamma(a+x+y)\Gamma(m-x+B+n-y)}{\Gamma(a+m+n+B)} \quad y \geq 0.$$

Modelo beta-binomial

$y|\theta \sim Bin(n, \theta) \quad \theta \sim Beta(a, b)$

$$p(\theta|y) \propto p(y|\theta)p(\theta) = \binom{n}{y} \theta^{y}(1-\theta)^{n-y} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1}(1-\theta)^{b-1}$$

$$\propto \theta^{y+a-1}(1-\theta)^{n+b-y-1}$$

$$\theta|y \sim Beta(\theta|a+y, b+n-y-1)$$

Distribución marginal

$$p(y) = \int p(y|\theta)p(\theta)\,d\theta$$

Modelo Gamma-Poisson

$s|\theta \sim Poisson(n\theta)$

$\theta \sim Gamma(a, b)$

$p(\theta|s) \propto p(s|\theta)p(\theta) \propto$

Modelo Poisson: $y|\theta \sim Bin(n,\theta)$ $x|\theta \sim Bin(m,\theta)$ $\theta \sim Beta(a,b)$

1. Predictiva posterior $y|x$

$$p(y|x) = \int p(y|\theta)\, p(\theta|x)\, d\theta \qquad \text{Posterior } \theta|x$$

$$p(\theta|x) \propto p(x|\theta)\, p(\theta) = \binom{m}{x} \theta^{x} (1-\theta)^{m-x} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1}(1-\theta)^{b-1} d\theta$$

$$\sim Beta(a+x, b+m-x)$$

$$p(y|x) = \int \binom{n}{y} \theta^{y}(1-\theta)^{n-y} \frac{\Gamma(a+x+b+m-x)}{\Gamma(a+x)\Gamma(b+m-x)} \theta^{a+x-1}(1-\theta)^{b+m-x-1}$$

$$\propto \binom{n}{y} B(a+x+y, b+m-x+n-y) \cdot \frac{1}{B(a+x, b+m-x)} \qquad y = 0,1,\cdots,n$$

$$= Y|X=x \sim \text{Beta-binomial}(n, a+x, b+m-x)$$

2. $y|\theta \sim Poisson(\theta)$

a. Previa de Jeffreys para θ.

$$\pi(\theta) \propto \sqrt{I(\theta)} \qquad l = y\log\theta - \theta - \log(y!) \qquad I(\theta) = \frac{1}{\theta}$$

$$\pi(\theta) \propto \sqrt{1/\theta} \qquad \int_0^{\infty} \theta^{-1/2} d\theta = \infty$$

b. Posterior Una gama es propia cuando $a > 0$

$$\pi(\theta|y) \propto p(y|\theta)\pi(\theta) \propto \theta^{y} e^{-\theta} \cdot \theta^{-1/2} = \theta^{y-1/2} e^{-\theta}$$

$$\theta|y \sim Gamma(y+1/2, 1)$$

$$y + \tfrac{1}{2} > 0 \qquad y \in \{0,1,\cdots\}$$

3. Estimador optimo

a. Perdida cuadrática, media posterior.

$$\underset{\tilde\theta}{\text{Minimizar}} \; E_{\theta|y}[(\theta-\tilde\theta)^2] = \int (\theta-\tilde\theta)^2 p(\theta|y)\, d\theta,$$

$$\theta^2 - 2\theta\tilde\theta + \tilde\theta^2 \qquad E[\theta^2|y] - 2\tilde\theta E[\theta|y] + \tilde\theta^2$$

$$\frac{d}{d\tilde\theta} = -2E[\theta|y] + 2\tilde\theta = 0 \qquad \tilde\theta = E[\theta|y]$$

b. Perdida absoluta- Mediana posterior

$$E_{\theta|y}[|\theta-\tilde\theta|] = \int |\theta-\tilde\theta|\, p(\theta|y)\, d\theta = \int_{-\infty}^{\tilde\theta} (\tilde\theta-\theta) p(\theta|y)\, d\theta + \int_{\tilde\theta}^{\infty} (\theta-\tilde\theta) p(\theta|y)\, d\theta$$

$$\frac{d}{d\tilde\theta} = \int_{-\infty}^{\tilde\theta} p(\theta|y)\, d\theta - \int_{\tilde\theta}^{\infty} p(\theta|y)\, d\theta, \quad -\int_{-\infty}^{\tilde\theta} p(\theta|y)\, d\theta = \int_{\tilde\theta}^{\infty} p(\theta|y)\, d\theta$$

$$-P(\theta \le \tilde\theta|y) = P(\theta \ge \tilde\theta|y) \qquad P(\theta \le \tilde\theta|y) = \frac{1}{2}$$

$$P(\theta \le \tilde\theta|y) = 1 - P(\theta \le \tilde\theta|y)$$

c. Relación riesgo bayesiano

$$R_F = R_F(\theta,\tilde\theta) = E_{y|\theta}[L(\theta,\tilde\theta(y))] \qquad R_B(\tilde\theta) = E_\theta[R_F(\theta,\tilde\theta)] = \int R_F(\theta,\tilde\theta) p(\theta)\, d\theta$$

$$R_B(\tilde\theta) = \int \left(\int L(\theta,\tilde\theta|y) p(y,\theta)\, dy \right) p(\theta)\, d\theta = \int \left(\int L(\theta,\tilde\theta(y)) p(\theta|y)\, d\theta \right) p(y)\, dy$$

$$= \int E_{\theta|y}(L(\theta,\tilde\theta(y)))\, p(y)\, dy = E_y[E_{\theta|y}(L(\theta,\tilde\theta(y)))]$$

4. Considere el modelo Gamma-Poisson

$y_i|\theta \sim \text{Poisson}(\theta)$ $\theta \sim \text{Gamma}(a,b)$ $Y=(y_1,\dots,y_n)$ $\sum_{i=1}^{n} y_i = n\bar{y}$

a. Estimador bayesiano $E[\theta|y]=\tilde{\theta}$

$$p(\theta|y) \propto \int p(y|\theta)p(\theta)d\theta = \int \prod_{i=1}^{n} p(y_i|\theta)\, p(\theta)\, d\theta = \int \prod_{i=1}^{n} \frac{e^{-\theta}\theta^{y_i}}{y_i!} \frac{b^a}{\Gamma(a)} \theta^{a-1} e^{-b\theta}\, d\theta$$

$$\propto \int \theta^{\sum y_i + a - 1} e^{-n\theta - b\theta} \sim \text{Gamma}(a+\textstyle\sum y_i,\ n+b)$$

Esperanza posterior $\tilde{\theta} = \frac{a+\sum y_i}{b+n} = \frac{a+n\bar{y}}{b+n}$ $\tilde{\theta} = \frac{a}{b+n} + \frac{n}{b+n}\bar{y}$ $\frac{a}{b+n} > 0$ $\frac{n}{b+n} \in (0,1)$.

b. Riesgo frecuentista $\tilde{\theta}_{MLE} = \bar{y}$ $E[\bar{y}]=\theta$ $E[y_i]=\theta$ $\text{Var}_\theta(y_i)=\theta$ $\text{Var}(\bar{Y})=\theta/n$

$R(\theta,\tilde{\theta}) = \text{Var}_\theta(\tilde{\theta}) + (\text{Sesgo}_\theta(\tilde{\theta}))^2$ $E[\tilde{\theta}] = E_\theta\left(\frac{a}{b+n} + \frac{n}{b+n}\bar{y}\right)$

Sesgo $\text{Bias} = E[\tilde{\theta}] - \theta = \frac{a}{b+n} + \frac{n}{b+n}\theta - \theta = \frac{a-b\theta}{b+n}$

Var $\text{Var}(\tilde{\theta}) = \left(\frac{n}{b+n}\right)^2 \frac{\theta}{n} = \frac{n\theta}{(b+n)^2}$

Riesgo $\frac{n\theta}{(b+n)^2} + \left(\frac{a-b\theta}{b+n}\right)^2$

Riesgo del estimador de máxima verosimilitud

$\tilde{\theta}_{MV} = \bar{Y}$ $E_\theta(\bar{Y})=\theta$ Insesgado $\text{Sesgo}(\bar{Y})=0$ $\text{Var}(\bar{Y}) = \frac{\theta}{n}$

$$R(\theta,\tilde{\theta}_{MV}) = \frac{\theta}{n}$$

c. Riesgo bayesiano $\theta \sim \text{Gamma}(a,b)$ $E[\theta]=a/b$ $\text{Var}(\theta) = \frac{a}{b^2}$

$$r(\tilde{\theta}) = E_\theta[R(\theta,\tilde{\theta})] = E_\theta\left(\frac{n\theta}{(b+n)^2} + \frac{(a-b\theta)^2}{(b+n)^2}\right) = \frac{n}{(b+n)^2}E(\theta) + \frac{1}{(b+n)^2}E[(a-b\theta)^2]$$

$$E[(a-b\theta)^2] = E(a^2 - 2ab\theta + b^2\theta^2) = b^2\,\text{Var}(\theta) = b^2 \cdot a/b^2 = a.$$

$$r(\tilde{\theta}) = \frac{a}{(b+n)^2}\left(\frac{n}{b}+1\right) = \frac{a(b+n)}{b(b+n)^2} = \frac{a}{b(b+n)}$$

5. Considere el modelo beta-binomial

$X|\theta \sim \text{Bin}(n,\theta)$ $\theta \sim \text{Beta}(\sqrt{n}/2, \sqrt{n}/2)$ Riesgo frecuentista

$$p(\theta|x) \propto p(x|\theta)p(\theta) \propto \theta^x (1-\theta)^{n-x}\, \theta^{\frac{\sqrt{n}}{2}-1} (1-\theta)^{\frac{\sqrt{n}}{2}-1}$$

$\theta|x \sim \text{Beta}\left(x+\frac{\sqrt{n}}{2},\ n+\frac{\sqrt{n}}{2}-x\right)$

$$R(\theta,\tilde{\theta}) = E_\theta[(\tilde{\theta}-\theta)^2] = E_\theta\left[\left(\tilde{\theta} - E_\theta(\tilde{\theta}) + E(\tilde{\theta}) - \theta\right)^2\right]$$
$$= E_\theta[(\tilde{\theta} - E_\theta(\tilde{\theta}))^2] + 2E[(\tilde{\theta}-E[\tilde{\theta}])(E[\tilde{\theta}]-\theta)] + E[(E(\tilde{\theta})-\tilde{\theta})^2]$$
$$= \text{Var}(\tilde{\theta}) + \text{Sesgo}(\tilde{\theta})^2$$

$\tilde{\theta} = E[\theta|x] = \frac{x+\sqrt{n}/2}{n+\sqrt{n}}$ $\text{Var}\left(\frac{x}{n+\sqrt{n}}\right) = \frac{n\theta}{(n+\sqrt{n})^2}$ $\text{Sesgo} = E[\tilde{\theta}] - \theta = \frac{\sqrt{n}/2 + n\theta}{n+\sqrt{n}} - \theta$.

$$R(\theta, \tilde{\theta}) = \mathrm{Var}(\tilde{\theta}) + \mathrm{Sesgo}(\tilde{\theta})^2 = \frac{n\theta(1-\theta)}{(n+\sqrt{n})^2} + \frac{n(1/2-\theta)^2}{(n+\sqrt{n})^2} = \frac{n}{(n+\sqrt{n})^2} \cdot \frac{1}{4}.$$

Modelo binomial:

1. Considere el modelo beta-binomial $y|\theta \sim Bin(n,\theta)$ $\theta \sim Beta(a,b)$ $y \in Y = \{0,\dots,n\}$

a. Distribución marginal de y $p(y|\theta) = \binom{n}{y}\theta^y(1-\theta)^{n-y}$ $p(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\theta^{a-1}(1-\theta)^{b-1}$

$$p(y) = \int_0^1 p(y|\theta)p(\theta)d\theta = \binom{n}{y}\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\int_0^1 \theta^{y+a-1}(1-\theta)^{n+b-y-1}d\theta = \binom{n}{y}\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\frac{\Gamma(y+a)\Gamma(n+b-y)}{\Gamma(a+b+n)}$$

b. Media y varianza marginal

Esperanza marginal $E(y) = E(E(y|\theta))$ $E(y|\theta) = n\theta$ $E(n\theta) = \frac{na}{a+b}$.

Varianza marginal ~~~

Promedio posterior de θ $\theta|y \sim Beta(a+y, b+n-y)$ $E(\theta|y) = \frac{a+y}{a+b+n}$

3. Galenshare.

4. Demuestre la familia exponencial

5. Sea $p(y|\theta) = c(\phi)h(y)\exp(\phi t(y))$

a. Derive a ambos lados de la ecuación:

$\int_y p(y|\phi)dy = 1$ con respecto a ϕ para demostrar que $E(t(y)|\phi) = \frac{c'(\phi)}{c(\phi)}$

$$\frac{d}{d\phi}\int_y p(y|\phi)dy = 0 \quad \int_y \frac{d}{d\phi}p(y|\phi)dy = 0 \quad \int [h(y)(c'(\phi)\exp(\phi t(y)) + c(\phi)t(y)\exp(\phi t(y)))]$$

$$\int p(y|\theta)\left(\frac{c'(\phi)}{c(\phi)} + t(y)\right)dy = 0 \quad \frac{c'(\phi)}{c(\phi)}\underbrace{\int p(y|\theta)dy}_{\text{Densidad 1.}} + \int t(y)p(y|\theta)dy$$

$$\frac{c'(\phi)}{c(\phi)} + E(t(y)|\phi) = 0 \qquad E(t(y)|\theta) = \frac{-c'(\phi)}{c(\phi)}$$

b. Calcular $\frac{d}{d\phi}p(\phi)$ $p(\phi) \propto c(\phi)^{n_0}\exp(n_0 t_0 \phi)$ $\log p(\phi) = n_0 \log c(\phi) + n_0 t_0 \phi + C$.

$$\frac{d}{d\phi} = n_0\frac{c'(\phi)}{c(\phi)} + n_0 t_0 = \frac{p'(\phi)}{p(\phi)} \qquad p'(\phi) = p(\phi)\left(\frac{n_0 c'(\phi)}{c(\phi)} + n_0 t_0\right) \qquad E\left[-\frac{c'(\phi)}{c(\phi)}\right] = t.$$

5. $\phi = g(\theta)$ $\theta = h(\phi)$ $p_\theta(\theta)$ $p_\phi(\phi) = p_\theta(h(\phi)) \times \left|\frac{dh}{d\phi}\right|$

a. $\theta \sim \text{Beta}(a,b)$ $\phi = \log\left(\frac{\theta}{1-\theta}\right)$ $p_\phi(\phi)$

$e^\phi = \frac{\theta}{1-\theta}$ $e^\phi - \theta e^\phi = \theta$ $e^\phi = \theta(1+e^\phi)$ $\frac{e^\phi}{1+e^\phi} = \theta = h$ $\frac{dh}{d\phi} = \frac{e^\phi}{(1+e^\phi)^2}$

$$p_\phi(\phi) = \frac{1}{B(a,b)}\left(\frac{e^\phi}{1+e^\phi}\right)^{a-1}\left(\frac{1}{1+e^\phi}\right)^{b-1}\frac{e^\phi}{(1+e^\phi)^2} = \frac{e^\phi}{(1+e^\phi)^2}$$

b. $\theta \sim \text{Gamma}(a,b)$ $\phi = \log\theta$ $\theta = e^\phi = h$ $e^\phi = \frac{dh}{d\phi}$ $\text{Gama}(a,b)$ $\frac{b^a}{\Gamma(a)}\theta^{a-1}e^{-b\theta}$

$$p_\phi(\phi) = \frac{b^a}{\Gamma(a)}e^{\phi(a-1)}e^{-b(e^\phi)}\cdot e^\phi = \frac{b^a}{\Gamma(a)}e^{a\phi - be^\phi}$$

6. Jeffreys $p_J(\theta) \propto \sqrt{I(\theta)}$ $I(\theta) = -E_{y|\theta}\left(\frac{d^2}{d\theta^2}\log p(y|\theta)\right)$ $n\,I = I_n$

a. $y|\theta \sim \text{Bin}(n,\theta)$

$p(y|\theta) = \binom{n}{y}\theta^y(1-\theta)^{n-y}$ $\log p(y|\theta) = y\log\theta + (n-y)\log(1-\theta)$

$\frac{d}{d\theta}\log p(y|\theta) = \frac{y}{\theta} - \frac{(n-y)}{1-\theta}$ $\frac{d^2}{d\theta^2}\log p(y|\theta) = -\frac{y}{\theta^2} - \frac{(n-y)}{(1-\theta)^2}$ $E(y) = n\theta$

$$-E_{y|\theta}\left(\frac{d^2}{d\theta^2}\log p(y|\theta)\right) = -\left(-\frac{n\theta}{\theta^2} - \frac{n-n\theta}{(1-\theta)^2}\right) = \frac{n}{\theta(1-\theta)}$$

b. Reparametrize la binomial $\psi = \text{logit}(\theta)$ $p(y|\psi) \propto e^{\psi y}(1+e^\psi)^{-n}$ $\binom{n}{y}\theta^y(1-\theta)^{n-y}$

$\psi = \log\left(\frac{\theta}{1-\theta}\right)$ $e^\psi = \frac{\theta}{1-\theta}$ $\theta = \frac{e^\psi}{1+e^\psi}$ $(1-\theta) = \frac{1}{1+e^\psi}$

$$p(y|\psi) = \binom{n}{y}\left(\frac{e^\psi}{1+e^\psi}\right)^y\left(\frac{1}{1+e^\psi}\right)^{n-y} = \binom{n}{y}\frac{e^{\psi y}}{(1+e^\psi)^n}$$

Información de Fisher de ψ

$$\ell(\psi) = \psi y - n\log(1+e^{\psi}) \qquad \frac{d\ell}{d\psi} = y - \frac{n e^{\psi}}{1+e^{\psi}} \qquad \frac{d\ell^2}{d\psi} = -n\left(\frac{(1+e^{\psi})e^{\psi} - e^{\psi}e^{\psi}}{e^{\psi 2}}\right)$$

$$\frac{d\ell^2}{d\psi} = \frac{-n\, e^{\psi}}{(1+e^{\psi})^2} \qquad I(\psi) = n\theta(1-\theta) \qquad \pi_J(\psi) \propto \sqrt{I(\psi)} = \sqrt{n\theta(1-\theta)}$$

$$\pi_J(\psi) \propto \frac{e^{\psi/2}}{1+e^{\psi}} \qquad I(\psi) = I(\theta)\left(\frac{d\theta}{d\psi}\right)^2 \quad \text{(transformación)}$$

7. $x|\theta \sim N(\theta, \theta)$, $\theta > 0$ $\qquad f(x|\theta) = \frac{1}{\sqrt{2\pi\theta}} \exp\left(\frac{-(x-\theta)^2}{2\theta}\right)$

$$\ell f(x|\theta) = \frac{1}{2}\log(2\pi\theta) + \left(\frac{-(x-\theta)^2}{2\theta}\right) = \frac{1}{2}\log(2\pi\theta) - \frac{(x-\theta)^2}{2\theta}$$

$$= -\frac{1}{2}\log(2\pi\theta) - \frac{x^2}{2\theta} + x - \frac{\theta}{2}$$

$$\ell'(\theta) = -\frac{1}{2\theta} + \frac{x^2}{2\theta^2} - \frac{1}{2} \qquad \ell''(\theta) = \frac{1}{2\theta^2} - \frac{x^2}{\theta^3} \qquad E[x^2] = Var(x) + E[x]^2 = \theta + \theta^2$$

$$I(\theta) = -E[\ell''(\theta)] = \frac{1}{2\theta^2} - \frac{(\theta+\theta^2)}{\theta^3} = -\frac{1}{2\theta^2} - \frac{1}{\theta} = \frac{2\theta+1}{2\theta^2}$$

$$p_J(\theta) \propto \sqrt{I(\theta)} = \sqrt{\frac{2\theta+1}{2\theta}}$$

8. MLE de θ y $J(\theta)/n$ $\qquad Y_i|\theta \overset{i.i.d}{\sim} Ber(\theta) \qquad p(y_i|\theta) = \theta^{y_i}(1-\theta)^{1-y_i}$

$$L(\theta|y) = \prod_{i=1}^{n} \theta^{y_i}(1-\theta)^{1-y_i} = \theta^{\sum y_i}(1-\theta)^{n-\sum y_i} \qquad \ell(\theta|y) = \sum y_i \log\theta + (n-\sum y_i)\log(1-\theta)$$

$$\ell'(\theta|y) = \frac{\sum y_i}{\theta} + \frac{(n-\sum y_i)}{1-\theta} \qquad \frac{\sum_i^n y_i}{\theta} = \frac{n-\sum y_i}{1-\theta} \qquad (1-\theta)\sum_{i=1}^{n} y_i = \theta(n-\sum y_i) \qquad \tilde{\theta} = \frac{\sum_{i=1}^{n} y_i}{n} = \bar{y}$$

$$J(\theta)/n \qquad J(\theta) = -\frac{d^2\ell(\theta|y)}{d\theta^2} = \frac{\sum y_i}{\theta^2} + \frac{n-\sum y_i}{(1-\theta)^2} \qquad \frac{J(\theta)}{n} = \frac{\bar{y}}{\theta^2} + \frac{1-\bar{y}}{(1-\theta)^2}$$

$$J(\tilde{\theta})/n = \frac{\bar{y}}{\bar{y}^2} + \frac{(1-\bar{y})}{(1-\bar{y})^2} = \frac{1}{\bar{y}} + \frac{1}{1-\bar{y}} \qquad \ell''(\theta|y) = -\frac{\sum y_i}{\theta^2} - \frac{n-\sum y_i}{(1-\theta)^2} < 0$$

$\sum y_i \geq 0$ $\quad n - \sum y_i \geq 0$ Denominador no negativos.

b. Encontrar $p_U(\theta)$ $\qquad \log p_U(\theta) = \frac{\ell(\theta|y)}{n} + c \qquad \ell(\theta|y) = (\sum y_i)\log\theta + (n-\sum y_i)\log(1-\theta)$

$$\frac{\ell(\theta|y)}{n} = \bar{y}\log\theta + (1-\bar{y})\log(1-\theta) + c. \qquad p_U \propto \theta^{\bar{y}}(1-\theta)^{1-\bar{y}}$$

c. Densidad proporcional

$$p(y_1, \dots, y_n|\theta)p_U(\theta)$$

$$p(y|\theta)p(\theta) \propto \theta^{\sum y_i + \bar{y}}(1-\theta)^{n-\sum y_i + 1 - \bar{y}}$$

Beta(a,b) $\qquad a-1 = (n+1)\bar{y} \qquad b-1 = (n+1)(1-\bar{y}) \qquad \theta \sim Beta((n+1)\bar{y}+1, (n+1)(1-\bar{y})+1)$

No es una prior bayesiana porque $\bar{y}$ utiliza

Intercambiabilidad:

- Considere el modelo $y_i|\theta \overset{iid}{\sim} N(\theta, \sigma_0^2)$, para $i=1,\dots,n$ con varianza σ_0^2 conocida, y $\theta \sim N(M_0, T_0^2)$ donde M_0 y T_0^2 son hiperparametros conocidos.

a. Hallar la distribución posterior de θ.

$$p(\theta|y) \propto p(y|\theta)p(\theta) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left[\frac{-(y_i-\theta_0)^2}{2\sigma_0^2}\right] \cdot \frac{1}{\sqrt{2\pi T_0^2}} \exp\left[\frac{(\theta-M_0)^2}{2T_0}\right]$$

$$\propto \exp\left[\frac{\sum -(y_i-\theta_0)^2}{2\sigma_0^2}\right] \exp\left[\frac{-\theta^2+2\theta M_0 - M_0^2}{2T_0}\right] = \exp\left[\frac{-\sum(y_i^2-2y_i\theta+\theta^2)}{2\sigma_0^2} + \frac{-\theta^2+2\theta M_0 - M_0^2}{2T_0}\right]$$

$$= \exp\left[-\frac{1}{2}\left[\sum_{i=1}^{n}(y_i-\theta)^2 + (\theta-M_0)^2\right]\right] \propto \exp\left[\frac{-\sum y_i^2}{2\sigma_0^2} + \frac{\sum 2y_i\theta}{2\sigma_0^2} - \frac{n\theta^2}{2\sigma_0^2} - \frac{1}{2T_0^2}\theta^2 + \frac{2M_0\theta}{2T_0^2} - \frac{M_0^2}{2T_0^2}\right]$$

$$\propto \exp\left\{\frac{2n\bar{y}\theta}{2\sigma_0^2} - \frac{n\theta^2}{2\sigma_0^2} - \frac{\theta^2}{2T_0^2} + \frac{2M_0\theta}{2T_0^2}\right\} = \exp\left\{-\frac{\theta^2}{2}\left(\frac{n}{\sigma_0^2}+\frac{1}{T_0^2}\right) + \theta\left(\frac{n\bar{y}}{\sigma_0^2}+\frac{M_0}{T_0^2}\right)\right\}$$

$$A = \frac{n}{\sigma_0^2}+\frac{1}{T_0^2} \qquad B = \frac{n\bar{y}}{\sigma_0^2}+\frac{M_0}{T_0^2} \qquad \exp\left\{-\frac{\theta^2}{2}A + \theta B\right\} \exp\left\{\frac{-A}{2}\left[\theta^2 - \frac{2B\theta}{A}\right]\right\}$$

$$\propto \exp\left\{\frac{-A}{2}\left(\theta-\frac{B}{A}\right)^2\right\} \qquad M_1 = \frac{B}{A} \qquad A = \frac{1}{T_0^2}$$

$$\theta|y \sim N\left(\frac{\frac{n\bar{y}}{\sigma_0^2}+\frac{M_0}{T_0^2}}{\frac{n}{\sigma_0^2}+\frac{1}{T_0^2}}, \left(\frac{n}{\sigma_0^2}+\frac{1}{T_0^2}\right)^{-1}\right)$$

b. Prevria de Jeffreys de θ.

$$-E_{y_i|\theta}\left[\frac{d^2}{d\theta^2}\log p(y_i|\theta)\right] = I(\theta)$$

$$p(y_i|\theta) = \frac{1}{\sqrt{2\pi\sigma_0^2}}\exp\left\{\frac{-(y_i-\theta)^2}{2\sigma_0^2}\right\} \qquad \log p(y_i|\theta) = \log\left[1/\sqrt{2\pi\sigma_0^2}\right] - \frac{(y_i-\theta)^2}{2\sigma_0^2} \propto -\frac{(y_i^2-2y_i\theta+\theta^2)}{2\sigma_0^2}$$

$$\frac{d\log p(y_i|\theta)}{d\theta} = \frac{2y_i-2\theta}{2\sigma_0^2} = \frac{y_i-\theta}{\sigma_0^2} \qquad \frac{d^2}{d\theta}\log p(y_i|\theta) = \frac{-1}{\sigma_0^2}$$

$$-E_{y_i|\theta}\left[\frac{-1}{\sigma_0^2}\right] = \frac{1}{\sigma_0^2} \qquad nI(\theta) = I_n(\theta)$$

$$\sqrt{\frac{n}{\sigma_0^2}}, \quad \pi_J(\theta) \propto 1.$$

n, σ_0^2 son conocidos.

- Considere el modelo jerarquico de la forma

$$y_i|\lambda \overset{iid}{\sim} Exp(\lambda), \text{ ie. } p(y_i|\lambda) = \frac{1}{\lambda}\exp\left(\frac{-y_i}{\lambda}\right) \text{ para } y_i>0 \quad i=1,\dots,n \quad \lambda>0$$

$$\lambda \sim Gam\text{-}inv(\alpha,\beta), \text{ ie. } p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{-(\alpha+1)}\exp\left(\frac{-\beta}{\lambda}\right) \quad \alpha>0 \quad \beta>0$$

a. Hallar la distribución predictiva posterior para m observaciones futuras $S^* = \sum_{i=1}^{m} y_i^*$

$$p(S^*|Y) = \int p(S^*|\lambda)p(\lambda|Y)$$

$$p(\lambda|y) \propto p(y|\lambda)p(\lambda) = \prod_{i=1}^{n} \frac{1}{\lambda} \exp\left(\frac{-y_i}{\lambda}\right) \left(\frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{-(\alpha+1)} \exp\left(\frac{-\beta}{\lambda}\right)\right)$$

$$\propto \lambda^{-n} \exp\left(\frac{-\sum y_i}{\lambda}\right) \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{-(\alpha+1)} \exp\left(\frac{-\beta}{\lambda}\right) \propto \exp\left(\frac{-\sum y_i - \beta}{\lambda}\right) \lambda^{-(\alpha+1)-n}$$

$$\sim \text{Gamma-inv}(\alpha+n,\ \textstyle\sum y_i + \beta)$$

$$p(\delta^*|y) = \int p(\delta^*|\lambda)\, p(\lambda|y)\, d\lambda \qquad \delta^*|\lambda \sim \text{Gamma}(m, \lambda)$$

$$\int \frac{\lambda^{-m}}{\Gamma(m)} \delta^{*m-1} e^{-\frac{\delta^*}{\lambda}} \frac{(\sum y_i+\beta)^{\alpha+n}}{\Gamma(\alpha+n)} \lambda^{-(\alpha+n+1)} \exp\left(\frac{-\sum y_i-\beta}{\lambda}\right) d\lambda \qquad M_X(t) = \frac{\lambda}{\lambda - t} \qquad M_{S_n}(t) = \prod_{i=1}^{n} M_X(t) = \left(\frac{\lambda}{\lambda-t}\right)^n$$

$$\text{Gamma}(n, \lambda).$$

$$\sum_{i=1}^{n} X_i \sim \Gamma(n, \lambda)$$

$$= \frac{(\delta^*)^{m-1}(\sum y_i+\beta)^{\alpha+n}}{\Gamma(m)\,\Gamma(\alpha+n)} \int \exp\left(-\frac{\delta^*}{\lambda} - \frac{\sum y_i - \beta}{\lambda}\right) \lambda^{-m-\alpha-n-1} d\lambda$$

Nucleo gamma.

$$= \frac{(\sum y_i+\beta)^{\alpha+n}(\delta^*)^{m-1}}{\Gamma(m)\,\Gamma(\alpha+n)} \cdot \frac{\Gamma(m+\alpha+n)}{(\sum y_i+\beta+\delta)^{m+\alpha+n}}$$

Ejercicios:

Exponencial + Gamma

$Y_i|\lambda \overset{iid}{\sim} \text{Exp}(\lambda)$ $\quad p(y_i|\lambda) = \frac{1}{\lambda} e^{-y_i/\lambda}$ $\quad \lambda \sim \text{Gamma-inv}(\alpha, \beta)$ $\quad p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{-(\alpha+1)} e^{-\beta/\lambda}$

a. $p(\lambda|y_1,\dots,y_n) \propto p(y_1\cdots y_n|\lambda)p(\lambda) = \prod p(y_i|\lambda)\, p(\lambda)$

$$= \prod \frac{1}{\lambda} \exp\left\{\frac{-y_i}{\lambda}\right\} \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{-(\alpha+1)} \exp\left\{\frac{-\beta}{\lambda}\right\}$$

$$\propto \lambda^{-(\alpha+n+1)} \exp\left\{\frac{-\sum y_i - \beta}{\lambda}\right\} \sim \text{Gamma-inv}\left(\alpha+n,\ \textstyle\sum y_i + \beta\right)$$

$$\alpha_0 = \alpha+n \qquad \beta_0 = \textstyle\sum y_i + \beta.$$

b. $p(Y^*|Y) = \int p(Y^*|\lambda)\, p(\lambda|y)$

$$= \int \frac{1}{\lambda} \exp\left(\frac{-Y^*}{\lambda}\right) \cdot \frac{\beta_0^{\alpha_0}}{\Gamma(\alpha_0)} \lambda^{-(\alpha_0+1)} e^{-\frac{\beta_0}{\lambda}}\, d\lambda.$$

$$= \frac{\beta_0^{\alpha_0}}{\Gamma(\alpha_0)} \int \lambda^{-(\alpha_0+1+1)} \exp\left(\frac{-(Y^*+\beta_0)}{\lambda}\right) d\lambda.$$

$$= \frac{\beta_0^{\alpha_0}}{\Gamma(\alpha_0)} \frac{\Gamma(\alpha_0+2)}{(Y^*+\beta_0)^{\alpha_0+2}} = \frac{(\sum y_i+\beta)^{\alpha+n}}{\Gamma(\alpha+n)} \frac{\Gamma(\alpha+n+1)}{(Y^*+\sum y_i+\beta)^{\alpha+n+1}}$$

Depende datos e hiperparametros.

c. $S^* = \sum_{i=1}^{m} Y_i^*$

$$p(s^*|y) = \int p(s^*|\lambda)\, p(\lambda|y)\, d\lambda.$$

$$= \int \frac{\lambda^{-m}}{\Gamma(m)} s^{*m-1} e^{-\frac{s^*}{\lambda}} \frac{B_0^{a_0}}{\Gamma(a_0)} \lambda^{-(a_0+1)} e^{-\frac{B_0}{\lambda}} d\lambda$$

$$= \frac{B_0^{a_0}}{\Gamma(a_0)\Gamma(m)} s^{*m-1} \int \lambda^{-m} \lambda^{-(a_0+1)} e^{\frac{-s^*}{\lambda}} e^{\frac{-B_0}{\lambda}} d\lambda$$

$$= \frac{B_0^{a_0}}{\Gamma(a_0)\Gamma(m)} s^{*m-1} \frac{\Gamma(a_0+m)}{(s^*+B_0)^{a_0+m}} = \frac{(\sum y_i + B)^{a+n}}{\Gamma(a+n)\Gamma(m)} s^{*m-1} \frac{\Gamma(a+n+m)}{(s^*+\sum y_i+B)^{a+n+m}}$$

$p(s^*|\lambda)$ es la suma de varias exponenciales por MGF

$M_Y(t) = \frac{\lambda}{\lambda - t}$

$M_{s^*} = \prod_{i=1}^{m} M_Y(t) = \left(\frac{\lambda}{\lambda+t}\right)^m$

MGF de una gamma.

$s^*|\lambda \sim \text{Gamma}(m, \lambda)$

$m=1$ $\quad \frac{(\sum y_i + B)^{a+n}}{\Gamma(a+n)\Gamma(1)} \frac{\Gamma(a+n+1)}{(y^* + \sum y_i + B)^{a+n+1}}$

Ejercicio 2: $Y_i|\lambda \overset{iid}{\sim} \text{Poisson}(\lambda)$ $\quad \lambda \sim \text{Gamma}(a, B)$ $\quad p(\lambda) = \frac{B^a}{\Gamma(a)} \lambda^{a-1} e^{-B\lambda}$

Posterior de λ

$$p(\lambda|Y) \propto p(Y|\lambda)p(\lambda)$$

$$\propto \prod_{i=1}^{n} p(y_i|\lambda)\, p(\lambda) = \prod_{i=1}^{n} \frac{\lambda^{y_i} e^{-\lambda}}{\lambda!} \cdot \frac{B^a}{\Gamma(a)} \lambda^{a-1} e^{-B\lambda}$$

$$\propto \lambda^{\sum y_i + a - 1} e^{-n\lambda} \cdot e^{-B\lambda} = \lambda^{\sum y_i} e^{-\lambda(n+B)}$$

$$\lambda|Y \sim \text{Gamma}(\sum y_i + a, n+B)$$

Posterior.

$$p(Y^*|Y) = \int p(Y^*|\lambda)\, p(\lambda|Y)$$

$$= \int \frac{\lambda^{y^*} e^{-\lambda}}{y^*!} \frac{(\sum y_i + a)^{n+B}}{\Gamma(\sum y_i + a)} \lambda^{\sum y_i + a - 1} e^{-\lambda(n+B)}$$

$$= \frac{(\sum y_i + a)^{n+B}}{y^*!\,\Gamma(\sum y_i + a)} \int \lambda^{\sum y_i + a + y^* - 1} e^{-\lambda(n+B+1)}$$

Nucleo de una gamma.

$$= \frac{(\sum y_i + a)^{n+B}}{y^*!\,\Gamma(\sum y_i + a)} \frac{\Gamma(\sum y_i + a + y^*)}{(n+B+1)^{\sum y_i + a + y^*}}$$

Para m observaciones. $\quad s^*|\lambda \sim \text{Poisson}(m\lambda)$ $\quad G_S(t) = \prod_{i=1}^{m} G_{Y_i}(t)$

$$= \prod_{i=1}^{m} \left(e^{\lambda(t-1)}\right) = e^{m\lambda(t-1)}.$$

$$p(s^*|Y) = \int p(s^*|\lambda)\, p(\lambda|Y)$$

$$= \int \frac{(m\lambda)^{s^*}}{s^*!} e^{-m\lambda} \frac{(\sum y_i + a)^{n+B}}{\Gamma(\sum y_i + a)} \lambda^{\sum y_i + a - 1} e^{-\lambda(n+B)} = \frac{(\sum y_i + a)^{n+B} m^{s^*}}{\Gamma(\sum y_i + a)\, s^*!} \int \lambda^{\sum y_i + a + s^* - 1} e^{-\lambda(n+B+m)}$$

$$= \frac{(\sum y_i + a)^{n+B} m^{s^*}}{\Gamma(\sum y_i + a)\, s^*} \frac{\Gamma(\sum y_i + a + s^*)}{(n+B+m)^{\sum y_i + a + s^*}}$$ ☑

3. Bernoulli - Beta $Y_i|p \overset{iid}{\sim} \text{Bernoulli}(p)$ $p \sim \text{Beta}(\alpha, \beta)$

Posterior $p(p|y) \propto p(y_i|p)p(p)$

$$\propto \prod_{i=1}^{n} (1-p)^{1-y_i} p^{y_i} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1}(1-p)^{\beta-1} = (1-p)^{n-\sum y_i} p^{\sum y_i} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1}(1-p)^{\beta-1}$$

$$\propto (1-p)^{n+\beta-\sum y_i-1} p^{\sum y_i+\alpha-1}$$

$$p|y \sim \text{Beta}\left(\sum y_i + \alpha,\ n+\beta-\sum y_i\right)$$

Predictiva posterior: Bernoulli $\longrightarrow$ Binomial.

$$p(y^*|y) = \int p(y^*|p)p(p|y)$$

$$=$$

M observaciones.

$$p(s^*|y) = \int p(s^*|p)\,p(p|y) \qquad S = \sum_{i=1}^{m} y_i \qquad y_i|p \sim \text{Bernoulli}(p)$$

$$= \int \binom{n}{x} p^s (1-p)^{n-s} \frac{\Gamma(}{\Gamma(\quad \Gamma(} \qquad S|p \sim \text{Binomial}(p)$$

Ejercicio 4: Normal con media conocida.

$$Y_i|M \overset{iid}{\sim} N(M, \sigma^2) \qquad M \sim N(M_0, \tau_0^2)$$

Posterior. $M|Y$

Jeffreys X
Transformación
Predictiva X
Posterior X

Transformaciones: